

Q.1)

1) Since $n=10$

$$\text{median} = (120 + 130) / 2 = 125 \text{ms}$$

$$2) \text{ Position} = \frac{90}{100} \times (n+1) = 0.9 \times 11 = 9.9$$

So, it is between 9th and 10th value

Interpolating:

$$\begin{aligned} P_{90} &= 200 + 0.9 \times (350 - 200) \\ &= 335 \text{ms} \end{aligned}$$

Discussion Answer

350ms is called an outlier, So, I was experienced slow response.

Q.2)

$$1) \text{ Position} = \frac{1}{4} (n+1) = 2.75$$

$$Q1 = 14 + 0.75 (15 - 14) = 14.75 \text{ms}$$

$$2) \text{ Position} = \frac{3}{4} (n+1) = 8.25$$

$$Q3 = 22 + 0.25 (25 - 22) = 22.75 \text{ms}$$

Q.3)

1) Position = $0.25 \times 11 = 2.75$

P25 = $40 + 0.75(42 - 40) = 41.5\%$

2) Position = $0.75 \times 11 = 8.25$

P75 = $70 + 0.25(80 - 70) = 72.5\%$

Q.4)

1) Number of values below 72 = 6

Percentile rank = $\frac{6}{10} \times 100 = 60\%$

2) The student scored better than 60% of students.

Discussion Answer

If a student is in the 70th percentile it means they performed better than 70% of the class and only 30% student scored higher
